

6 points

$$e) a) \Rightarrow [I_2] = f(t) - (0,25)$$

$$\text{courbe b)} \Rightarrow [I^-] = f(t) - (0,25)$$

$$2) a) [I^-]_{\infty} = 0,01 \text{ mol/L} \neq 0 \text{ donc}$$

 I^- est le réactif en excès. - (0,5)

$$b) [I^-]_0 = 0,05 \text{ mol/L} - (0,5)$$

$$[I^-]_f = 0,01 \text{ mol/L} - (0,5)$$

$$c) [H_2O_2]_0 = \frac{[I_2]_f}{1} = 0,02 \text{ mol/L} - (0,5)$$

$$d) [H_2O_2]_0 = \frac{C_1 V_1}{V_1 + V_2}$$

$$C_1 = \frac{[H_2O_2]_0 (V_1 + V_2)}{V_1} = 4 \cdot 10^{-2} \text{ mol/L} - (0,5)$$

$$C_2 = \frac{[I^-]_0 (V_1 + V_2)}{V_2} = 0,1 \text{ mol/L} - (0,5)$$

$$3) a) v_{t=10 \text{ min}}^{(I^-)} = 1,44 \cdot 10^{-3} \text{ mol/L} \cdot \text{min}^{-1} - (0,5)$$

$$v_{t=10 \text{ min}}^{(I_2)} = \frac{v_{t=10 \text{ min}}^{(I^-)}}{2} = 7,2 \cdot 10^{-4} \text{ mol/L} \cdot \text{min}^{-1} - (0,5)$$

b) la vitesse diminue au cours du temps car les quantités des réactifs diminuent. - (0,5)

$$4) a) \text{ A } t = t_{1/2} \Rightarrow [H_2O_2] = \frac{[H_2O_2]_0}{2} = 0,01 \text{ mol/L}$$

$$\Rightarrow [H_2O_2] = [I_2] = 0,01 \text{ mol/L}$$

$$t_{1/2} = 8 \text{ min} - (0,5)$$

$$b) i) t_{1/2} \text{ augmente} - (0,25)$$

$$ii) t_{1/2} \text{ diminue} - (0,25)$$

Exo 2 (8 points)

$$1) a) (m_0 + M) OG = m_0 R + M \cdot 0$$

$$11 M_0 OG = m_0 R$$

$$OG = \frac{m_0 R}{11 M} = \frac{m_0 R}{11 \times \frac{m_0}{10}} = \frac{10 R}{11} - (0,5)$$

$$b) \mu(\vec{r}) + \mu(\vec{r}') = J_D \ddot{\theta}$$

$$- \sin \theta \times m g OG + 0 = J_D \ddot{\theta}$$

$$J_D \ddot{\theta} + 11 M g \times \frac{10 R \sin \theta}{11} = 0$$

$$J_D \ddot{\theta} + 10 M g R \theta = 0$$

$$J_D = \frac{1}{2} M R^2 + m_0 R^2 = \frac{21 M R^2}{2}$$

$$\ddot{\theta} + \frac{20 g}{21 R} \theta = 0 - (1)$$

$$T = 2\pi \sqrt{\frac{21 R}{20 g}} = 0,92 \text{ s} - (0,5)$$

2) a)

$$E_m = \frac{1}{2} J_D \dot{\theta}^2 + 11 M g OG (1 - \cos \theta)$$

$$\frac{dE_m}{dt} = J_D \dot{\theta} \ddot{\theta} + 11 M g OG \sin \theta \dot{\theta} = 0$$

$$\dot{\theta} (J_D \ddot{\theta} + 11 M g OG \sin \theta) = 0$$

$$\dot{\theta} \neq 0 \Rightarrow J_D \ddot{\theta} + 11 M g OG \sin \theta = 0$$

$$\sin \theta \approx \theta$$

$$J_D \ddot{\theta} + 10 M g R \theta = 0 - (1)$$

$$\Rightarrow \ddot{\theta} + \frac{20 g}{21 R} \theta = 0$$

$$b) T = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{21 R}{20 g}}$$

$$\Rightarrow \frac{l}{g} = \frac{21 R}{20 g} \Rightarrow l = \frac{21 R}{20}$$

$$\text{AN: } l = \frac{21 \times 20}{20} = 21 \text{ cm} - (0,5)$$



$$\vec{r} + \vec{R} + \vec{f} = m_0 \vec{a}$$

$$f = m_0 \ddot{x} \Rightarrow -kx = m_0 \ddot{x}$$

$$\Rightarrow \ddot{x} + \frac{k}{m_0} x = 0$$

$$\ddot{x} + \frac{1}{\lambda} \dot{x} = 0$$

$$\frac{dv}{dt} + \frac{1}{\lambda} v = 0 \quad (1)$$

$$b) v = e \cdot \dot{e}$$

$$\frac{dv}{v} = -\frac{dt}{\lambda} \Rightarrow \ln v = -\frac{t}{\lambda} + C$$

$$v(t) = e^{-\frac{t}{\lambda} + C} = C_0 e^{-\frac{t}{\lambda}}$$

$$A t=0 \Rightarrow v = v_0$$

$$v_0 = C_0 e^0 \text{ d'où } C_0 = v_0$$

$$v(t) = v_0 e^{-\frac{t}{\lambda}} \quad (1)$$

$$c) \lim_{t \rightarrow \infty} v(t) = 0 \quad (0,5)$$

$$4) x(t) = \int v_0 e^{-\frac{t}{\lambda}} dt$$

$$x = -v_0 \lambda e^{-\frac{t}{\lambda}} + C$$

$$A t=0 \Rightarrow x=0$$

$$0 = -v_0 \lambda + C \Rightarrow C = \lambda v_0$$

$$x(t) = \lambda v_0 \left(1 - e^{-\frac{t}{\lambda}}\right) \quad (1)$$

$$b) d = \lim_{t \rightarrow \infty} x(t) = \lambda v_0 \quad (0,5)$$

$$d = \frac{m_0 v_0}{k} = \frac{2 \times 2}{4 \cdot 10^{-3}} = 1000 \text{ m} \quad (0,5)$$

Cours à domicile: 775136349 (6 points)

$$1) \text{ Mtq } v = \sqrt{\frac{GM_S}{a}} \quad (1)$$

$$\omega = \frac{v}{r} = \sqrt{\frac{GM_S}{a^3}} \quad (1)$$

2) la Terre, SOHO et le soleil restent constamment alignés, donc

$$\omega_T = \omega_H = \sqrt{\frac{GM_S}{a^3}} \quad (1)$$

3) SOHO $\Rightarrow$ MCU dans le référentiel héliocentrique.

$$\vec{a} = \omega^2 (a-x) \vec{n}$$

$$\vec{a} = \frac{GM_S}{a^3} (a-x) \vec{n}$$

$$\vec{a} = \frac{GM_T K}{a^3} (a-x) \vec{n} \quad (1)$$

$$4) \vec{F}_{T \rightarrow \text{sat}} + \vec{F}_{S \rightarrow \text{sat}} = m \vec{a}$$

$$- \frac{G m_T m}{x^2} \vec{n} + \frac{G K m_T m}{(a-x)^2} \vec{n}$$

$$= m \times \frac{G K m_T}{a^3} (a-x) \vec{n}$$

$$-\frac{1}{x^2} + \frac{K}{(a-x)^2} = \frac{K}{a^3} (a-x) \quad (1)$$

5) En tenant compte de l'approximation $\frac{x}{a} \ll 1$

$$\frac{K}{(a-x)^2} = \frac{K}{a^2 \left(1 - \frac{x}{a}\right)^2} = \frac{K}{a^2} \left(1 - \frac{x}{a}\right)^{-2}$$

$$\frac{K}{(a-x)^2} \approx \frac{K}{a^2} \left(1 + \frac{2x}{a}\right)$$

$$-\frac{1}{x^2} + \frac{K}{a^2} \left(1 + \frac{2x}{a}\right) = \frac{K}{a^3} (a-x)$$

$$-\frac{1}{x^2} + \frac{K}{a^2} + \frac{2Kx}{a^3} = \frac{K}{a^2} - \frac{Kx}{a^3}$$

$$\frac{3Kx}{a^3} = \frac{1}{x^2} \Rightarrow x^3 = \frac{a^3}{3K}$$

$$\frac{a}{\sqrt[3]{3K}} = 1,495 \cdot 10^9 \text{ m}$$